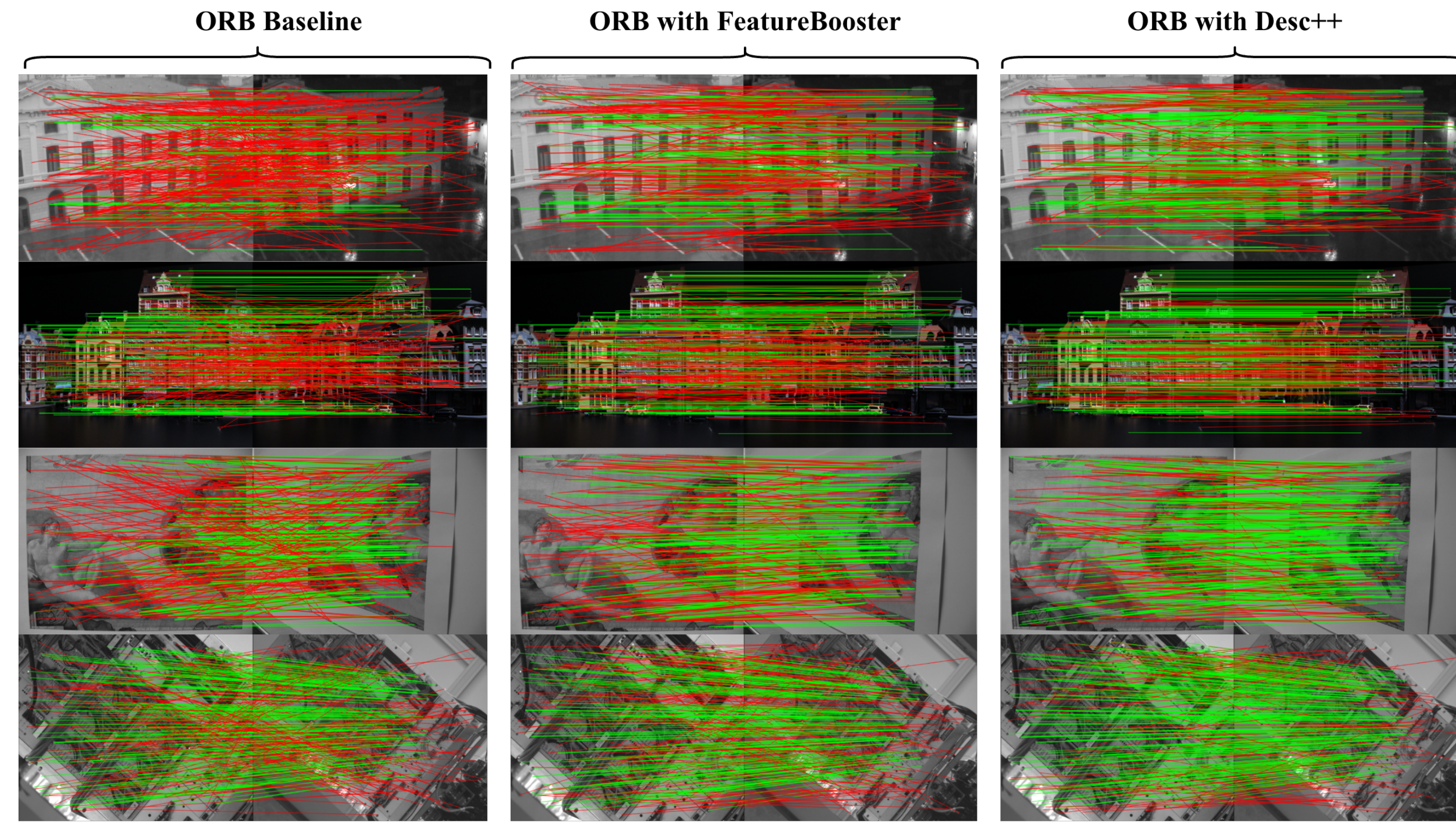


Desc++: Efficient Descriptor Enhancement for Data Association in Existing Visual SLAM Systems

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Motivation

Reliable **visual data association** determines the pose accuracy and map consistency of visual SLAM (V-SLAM). However:

- **Handcrafted descriptors** (e.g., ORB) used by mature real-time systems degrade under illumination and viewpoint changes.
- **Learned front-ends** (e.g., SuperPoint + LightGlue) are robust but require replacing the whole pipeline and are too heavy for embedded platforms.
- **Descriptor enhancement** refines existing descriptors in-place, but current methods use simplified attention with limited contextual modeling.

⇒ A fundamental **trade-off between contextual expressiveness and efficiency**.

What is Desc++?

A **lightweight, plug-and-play** enhancement module that jointly encodes descriptors and keypoint geometry, and aggregates spatial context via a **hybrid Mamba-AFT architecture in linear time** — enhanced descriptors keep the **same dimensionality and matching interface**, so they drop directly into deployed V-SLAM systems.

Contributions

- Hybrid context aggregation: order-agnostic global attention (AFT-Simple), and geometry-aware sequential modeling (Mamba) over Z-order serialized keypoints.
- Genuine plug-and-play: the same module integrated into **4 heterogeneous SLAM systems** without touching downstream logic.
- Analysis of *when and why* enhancement works: more tracked map points and tighter geometric constraints at a fraction of the cost of learned front-ends.

Problem Formulation

Given keypoint geometries $\mathcal{K} = \{k_i\}$ and descriptors $\mathcal{D} = \{d_i \in \mathbb{R}^D\}$, learn a lightweight mapping

$$f_\theta : (\mathcal{D}, \mathcal{K}) \rightarrow \hat{\mathcal{D}},$$

where the enhanced $\hat{d}_i$ retains the same dimension $\mathcal{D}$ while being more discriminative and stable, and can **directly replace** $\mathcal{D}$ in SLAM and matching pipelines (Fig. 1).

The Desc++ Architecture

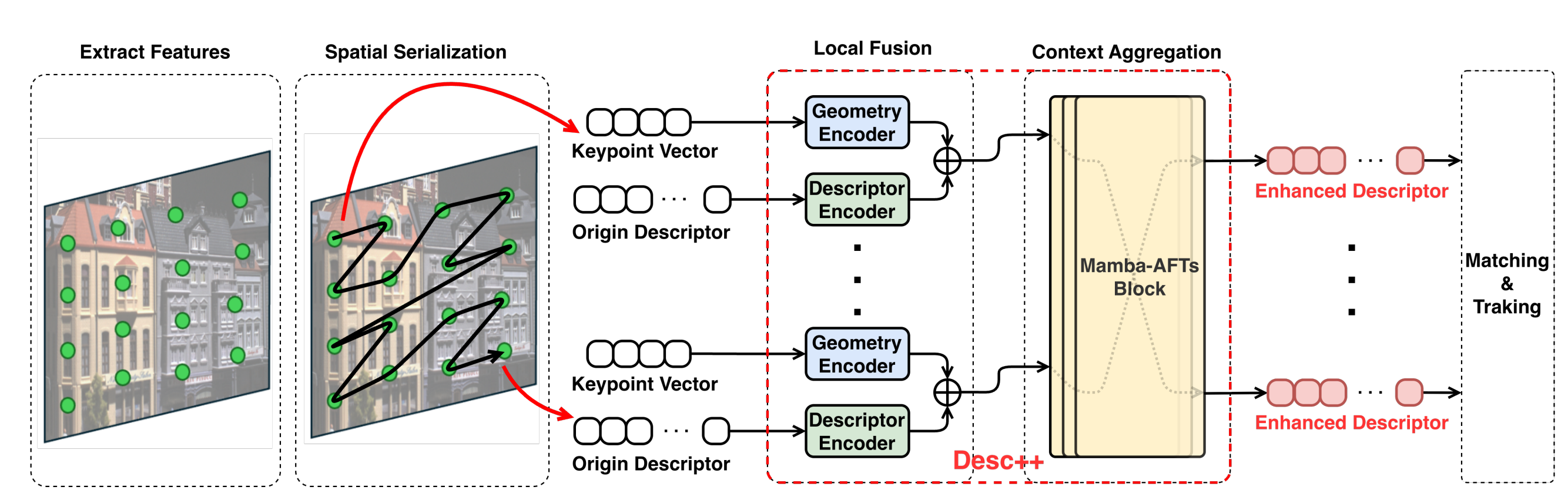


Figure 1. Overall pipeline: Z-order serialization → Local Fusion → Context Aggregation (Mamba-AFTs).

1. Z-order Spatial Serialization

Keypoints are sorted by Morton codes so that 2D neighbors stay adjacent in the 1D sequence — preserving spatial locality for the causal SSM.

2. Local Fusion Stage

Descriptor and geometry are encoded and fused:

$$\mathbf{z}_i = \text{MLP}_{\text{desc}}(d_i) + \text{MLP}_{\text{geo}}(\Phi_{\text{LFF}}(k_i))$$

with Learnable Fourier Features for fine-grained spatial structure.

3. Context Aggregation: Mamba-AFTs Block

Two parallel branches with complementary strengths, fused by a learned per-keypoint gate ($w_{\text{AFT},i} + w_{\text{Mamba},i} = 1$):

- **AFT-Simple branch**: permutation-invariant *global* context in $\mathcal{O}(ND)$ — no quadratic attention.
- **Mamba branch**: selective-scan SSM captures *sequential, geometry-aware* dependencies along the Z-order path.

Training

Trained on MegaDepth with a FastAP-based ranking objective:

$$\mathcal{L}_{\text{total}} = \underbrace{\frac{1}{N} \sum_i (1 - \text{AP}(\hat{d}_i))}_{\text{matching}} + \underbrace{\frac{1}{N} \sum_i \max(0, \frac{\text{AP}(d_i)}{\text{AP}(\hat{d}_i)} - 1)}_{\text{booster}}$$

$\mathcal{L}_{\text{match}}$ maximizes the matching AP of enhanced descriptors, while $\mathcal{L}_{\text{boost}}$ ensures they never fall below the originals. Only **~2.4M parameters**; ~2 days on one RTX 4090.

Descriptor Matching (HPatches)

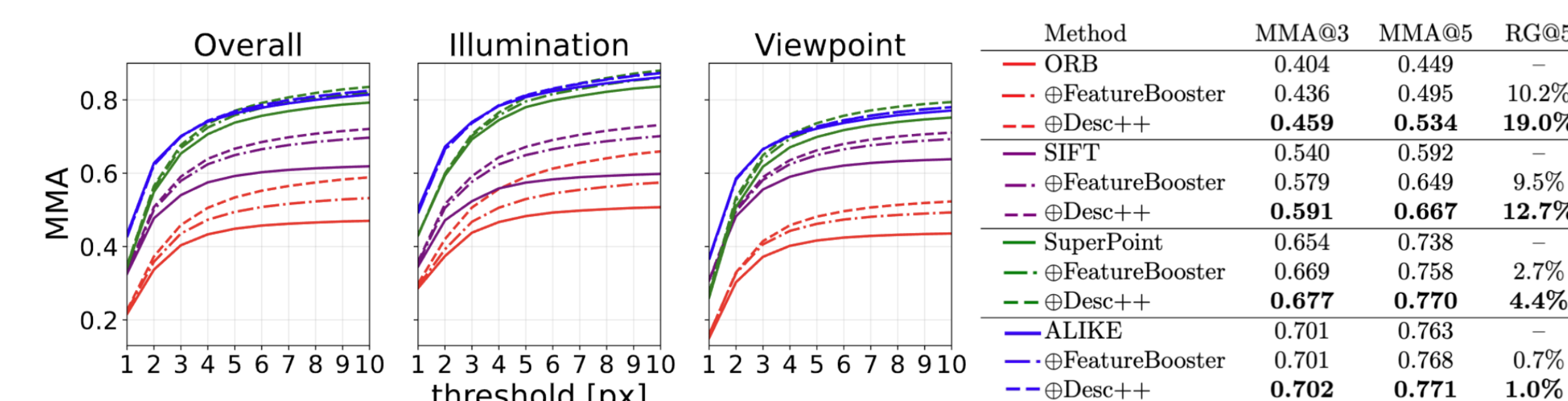


Figure 2. MMA curves: Desc++ vs. original descriptors and FeatureBooster.

Relative gains @5px (Fig. 2): **ORB 19.0%, SIFT 12.7%** (vs. 10.2% / 9.5% for FeatureBooster); consistent gains also on SuperPoint and ALIKE.

System-Level SLAM Results

Integrated *unchanged* into ORB-SLAM2, ORB-SLAM3, RGB-L, and MAVIS-SLAM. Quantitative results are summarized in Table 1, with qualitative examples shown in Fig. 3.

Table 1. System-level ATE RMSE summary (5 runs per sequence). “Improved” counts sequences with lower error than the unmodified base system.

Benchmark	Base system	Improved	Max rel. gain
EuRoC (stereo)	ORB-SLAM2	10/10	10.4%
EuRoC (stereo-inertial)	ORB-SLAM3	9/11	23.6%
KITTI (stereo-LiDAR)	RGB-L	10/11	37.2%
Hilti 2023 (multi-cam VI)	MAVIS-SLAM	9/9	37.3%

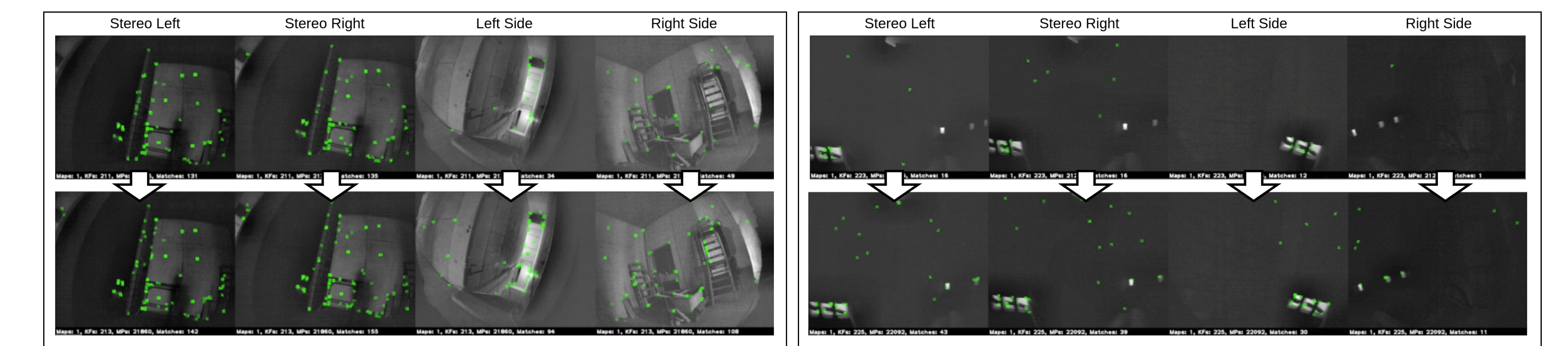


Figure 3. Qualitative feature correspondences on the Hilti sequences. Desc++ preserves more valid feature-to-map associations under large cross-camera viewpoint changes.

Accuracy-Efficiency Trade-off

Front-end	Latency (ms)	GPU Mem.	FPS
ORB-SLAM3	27.2	—	37
+ Desc++	32.5	544 MiB	30
SP + LightGlue	71.9	6130 MiB	14

Table 2. Only +5.3 ms over the baseline; ~11× less memory than a replacement-based learned front-end. Real-time(>30 FPS, 1k–2k keypoints).

Takeaways

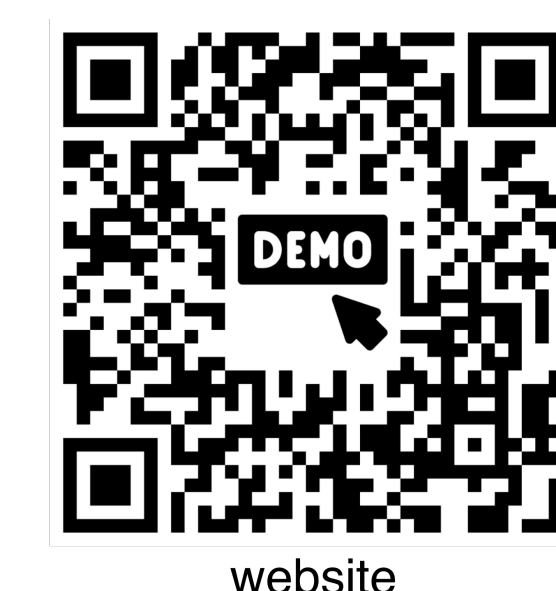
- Refining descriptor quality is a practical, efficient path to improving deployed real-time V-SLAM — no front-end redesign.
- Effective when features are degraded yet detectable; bounded by extraction failures (e.g., severe motion blur).
- Future work: joint per-descriptor confidence prediction for the SLAM back-end.

Selected References

[1] X. Wang *et al.*, “FeatureBooster,” CVPR 2023. [2] A. Gu and T. Dao, “Mamba,” arXiv 2023. [3] S. Zhai *et al.*, “An Attention-Free Transformer,” arXiv 2021. [4] C. Campos *et al.*, “ORB-SLAM3,” T-RO 2021.



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